

IDENTIFICATION

Product Code:	DEC-12-ZR5B-D
Product Name:	DIAL-MS BUILD Program Description
Date Created:	December 1, 1970
Maintainer:	Software Services

LAP6-DIAL is an editor, filing system and assembler for use with the PDP-12 computer. The editor and filing portions are derived from the basic LINC program LAP6¹ by Mary Allen Wilkes of Washington University. The assembly portion is derived from several programs used for the PDP-8 computer including PAL-D².

The Digital Equipment Corporation wishes to express to the author, Mary Allen Wilkes (Clark), and the Computer Research Laboratory of Washington University, St. Louis, Missouri, its appreciation for the development set forth in LAP6 as well as its thanks for permission to use parts of the LAP6 program.

¹M. A. Wilkes, LAP6 Handbook, Computer Research Laboratory Tech. Rep. No. 2, Washington University, St. Louis, May 1, 1967.

²PAL-D Assembler Programmer's Reference Manual DEC-D8-ASAA-D.

1.0 PROGRAM OVERVIEW

BUILD is the name of the routine which "customizes" DIAL-MS for a given configuration. BUILD determines what devices a user has and creates a set of I/O routines for DIAL-MS¹ to use. At present, BUILD resides in blocks 310, 345, 365, and 366 of the DIAL-MS system.

BUILD is composed of three separate logical components. The first is the actual code that decides which devices are present and constructs the necessary tables for the I/O controller to use. The second component is the I/O controller which dispatches the calls for operations to the correct handlers with any necessary modifications made in the data. The individual I/O handlers are the third logical part. Each I/O handler operates just one device, and each is independent of the controller and other routines. The three components of BUILD are described in the following section.

2. INITIALIZATION

The initialization phase resides at block 310 on the DIAL-MS system. Note that this is part of the Editor (locations 0-377). The DIAL-MS Editor requires that the pointers to the read and write routines reside at locations 21 and 22, respectively. Therefore, while the initialization phase starts at location 4020 in LINCmode it immediately jumps past the pointers at 21 and 22. It then reads in the prototype I/O controller located at block 345, plus all the handlers for all the possible devices (LINCtape and RS08, DF32 and RK08 disks) which are located at blocks 365 and 366 of the DIAL-MS tape. Initially the handlers and controllers are set up to support only LINCtape, with the LINCtape handler residing at 7600 of field 1. The program then checks for the various devices by device IOT's. If the device is present, the program alters the device table slightly to include these new unit codes. It will also move the new secondary device handler to location 7400 of field 1 if another device in addition to LINCtape is present. Locations 7400-7777 of field 1 are then written onto tape 0. Thus, the system now knows that blocks 322 and 323 of LINCtape 0 contain all the necessary information and they do not have to be rebuilt each time. Blocks 300-467 of tape 0 (with the exception of the index) are copied to blocks 0-167 of logical unit 100, providing 100 is not on

¹LAP6-DIAL-MS is commonly referred to as DIAL-MS.

LINCtape (logical unit 100 is discussed in the I/O controller section). The system then does a standard bootstrap of the DIAL-MS Editor.

3. I/O CONTROLLER

The I/O controller is essentially the dispatcher for DIAL-MS. Calls are made to read or write operations, and the controller passes this information on to the correct handler. The controller is a field independent routine (may be located in any field and called from any field). The calling sequences are discussed at length in the DIAL-MS manual. Inside the controller, there exists a table located at 7300 which contains the values of the units that are available, where the routines are to do the operations, and the amount to add (or subtract) from the block numbers when going from one device to another. This table is commonly referred to as the MTABLE. Each unit is given three words in this table. The first word is the actual unit number. The second is the calling address of the read handler for this unit. The last word is the number of blocks to add onto the user's calling argument to transform it correctly to the physical block number on the device. A minus unit number indicates the end of the table. In prototype form, the table contains eleven units. Eight of these are the LINCtapes (units 0-7); first word contains the unit number, second word contains 7630, the read routine entry point, the third word contains 0 because there is no correction necessary. In addition, there are three more units; namely, 100, 110 and 111. 100 is used internally to describe the DIAL-MS system. Thus, if you are running a tape system, DIAL-MS resides on blocks 300-367 of unit 0. The Editor resides on block 0 of unit 100 so the entry point in the table is 100 (unit #100), 7630 (pointer to the LINCtape routine), 300 (number of blocks to add to unit 100 calls to make it come out to the correct LINCtape blocks). This is how DIAL-MS can work off any device. The system is always on device 100; the MTABLE tells it where the device and block number really are. Unit 110 is the source unit; its first block is also 0. Thus, to reference block 3 of the source, it is block 3 of unit 110. The MTABLE tells the controller how to transform these numbers into physical device calls. Unit 111 is the binary unit (the first block of it is also 0). Only these three logical units are presently implemented; more can be added without any trouble.

The disk units are 10-17, with the pointer to 7430 (this is where the disk routine will be). Note that units 20-45 do not appear in the table if the user has multiple RK08s. The controller will look for units 10-17 if any unit between 20 and 47 is given as an argument to the controller. Note that it is trivial to make the values for the disk units those for the tape units, and vice versa. All that is necessary is to change the pointers to the handler locations. At present, only two handlers are in core at any one time (LINtape plus perhaps a disk handler). This can easily be changed to permit more handlers to be present if a user has a special device he wants to use.

An important point of interest about the I/O controller is that the controller leaves the read address of the handler just used at location 7770 and the number of blocks to add to any calling sequence in location 7771. This is necessary because some program (e.g., the Loader) may destroy the I/O controller, and then call the handlers directly, rather than going through the controller.

The only other routines not discussed are the bootstrap and the MOVE routines. The bootstrap merely reads in blocks 300-307 of the Editor off device 100 and jumps to 4021, instead of to 4020. Jumping to 4020 would cause the system to reload the I/O routines. The MOVE routine is so trivial that it will not be discussed here.

4. I/O HANDLERS

The I/O handlers are the low level routines of DIAL-MS. They control the physical devices. Though they may vary greatly in design and structure, they all have several things in common:

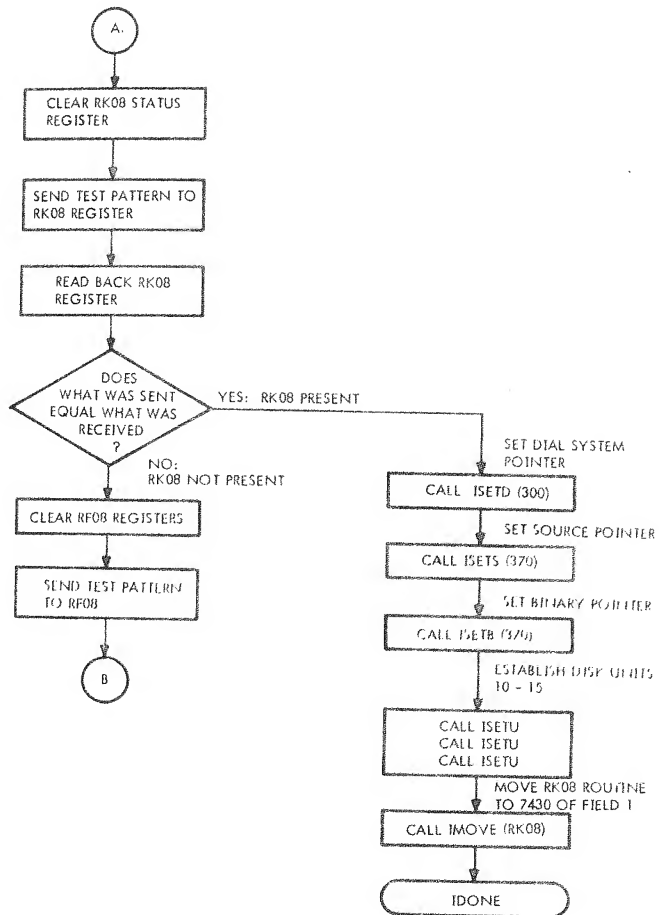
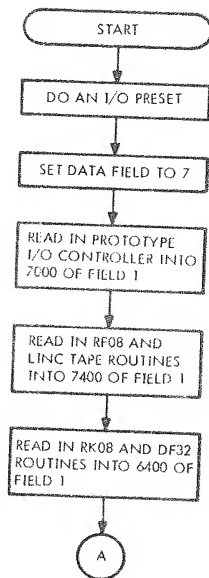
- a. They are all page independent (can be randomly located on any page in core and still function).
- b. No outside routines or constants are needed.
- c. They are field independent, but may be called only from the field in which they presently reside.
- d. Only location 30 through 167 of the page may be used by them.
- e. The read entry point is at 30 of the page and the write entry point is at 32 of the page.
- f. The calling sequence is similar to the calling sequence to the I/O controller, but the arguments are in-line, rather than pointed to by a pointer.

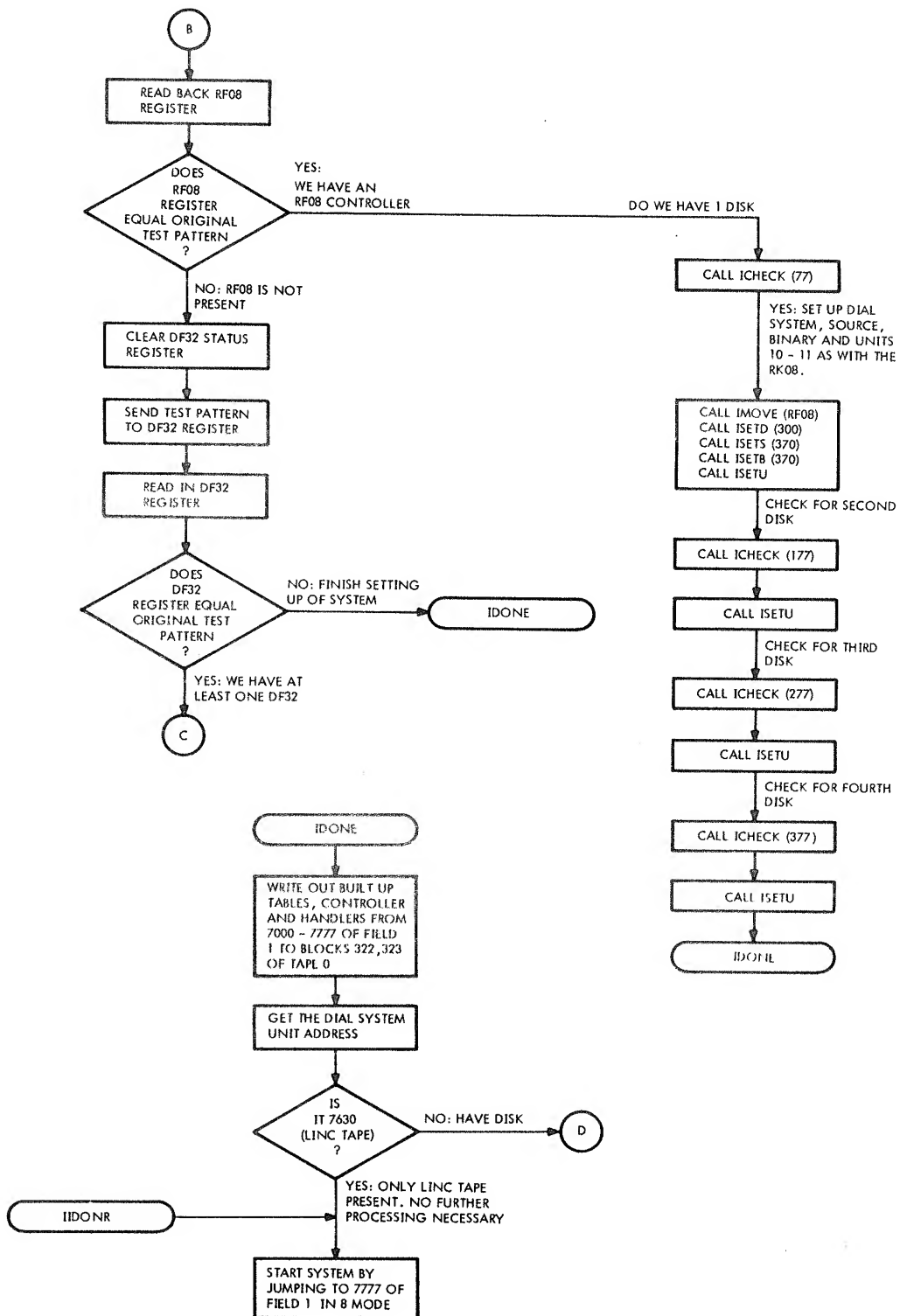
- g. There is no code at 150 or 151 of the page so that if they are ever located in 7600, no code will be over the disk data break locations.

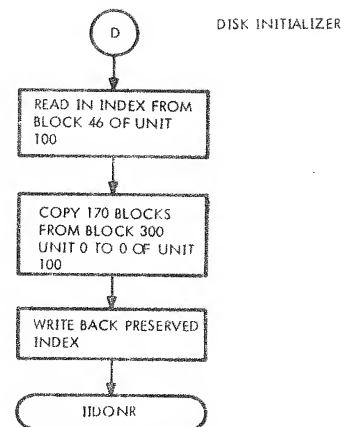
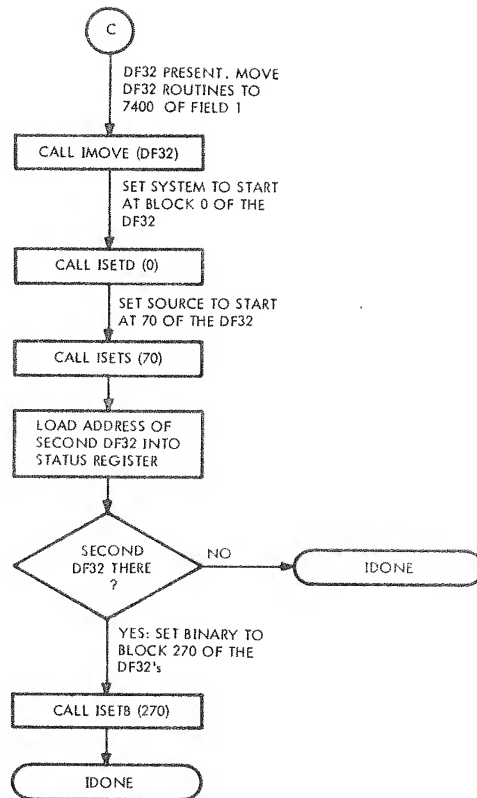
The I/O handlers also share one other feature. They only look at the rightmost digit of the calling unit, thus insuring the device independence of the system. While the physical nature of each handler is important, only the flow charts are necessary to understand each one. They are basically straightforward code. At present, there are four handlers implemented: LINCtape and RF08, DF32, and RK08, and disks. Any other device may be supported by merely writing a handler for it which conforms to the above standards.

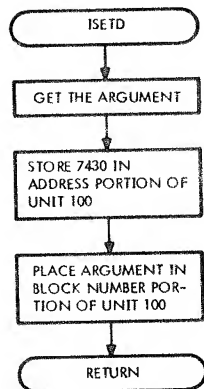
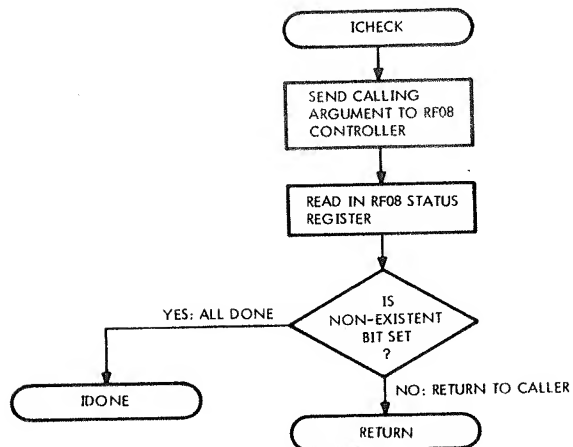
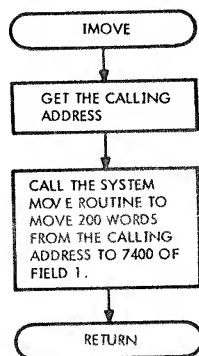
5.0 FLOW DIAGRAM (Attached)

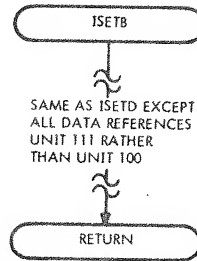
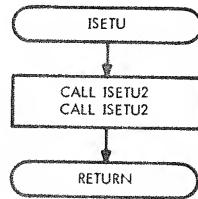
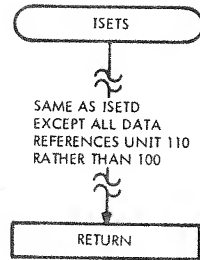
6.0 PROGRAM LISTING (Attached)



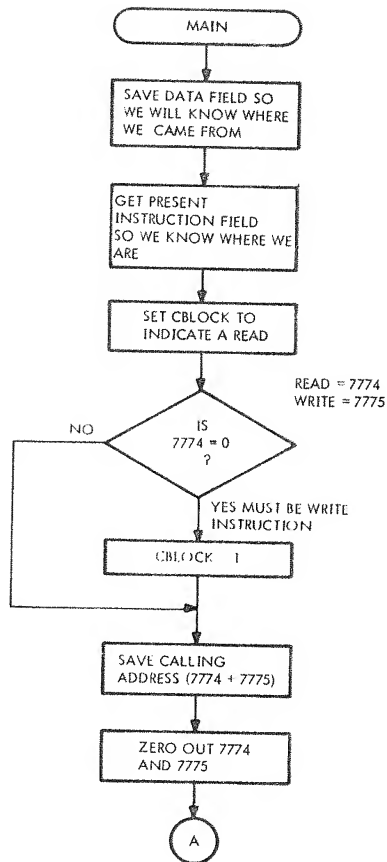
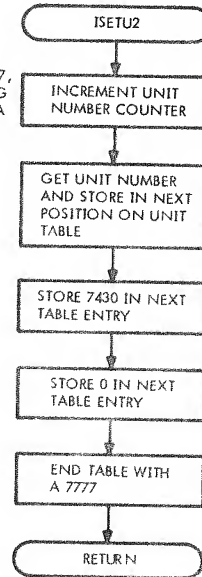


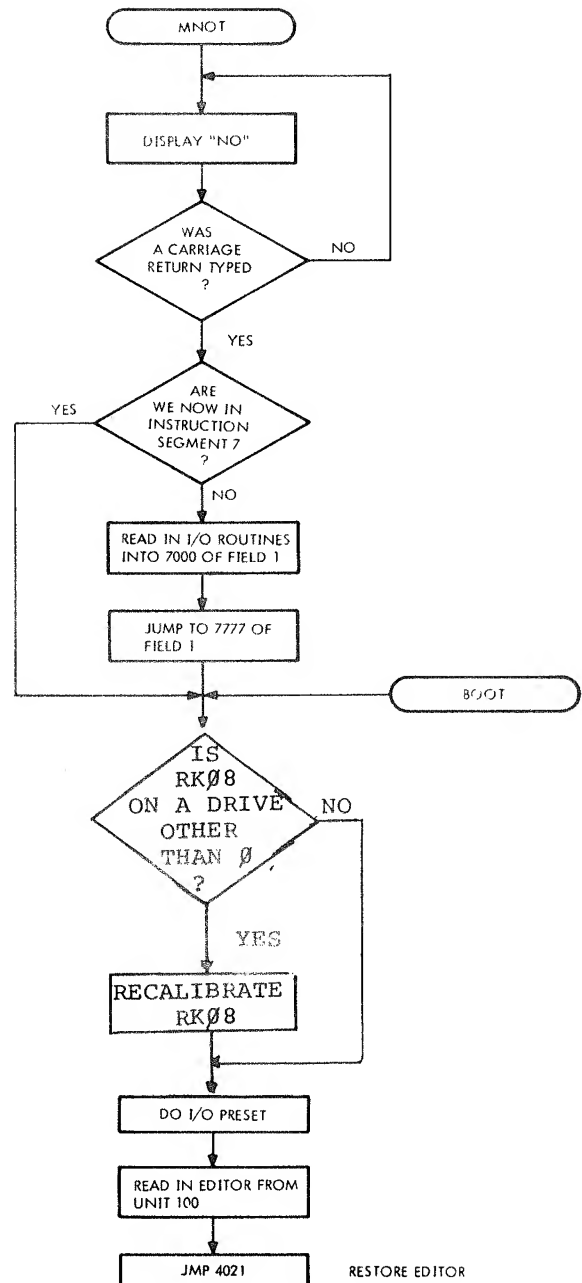
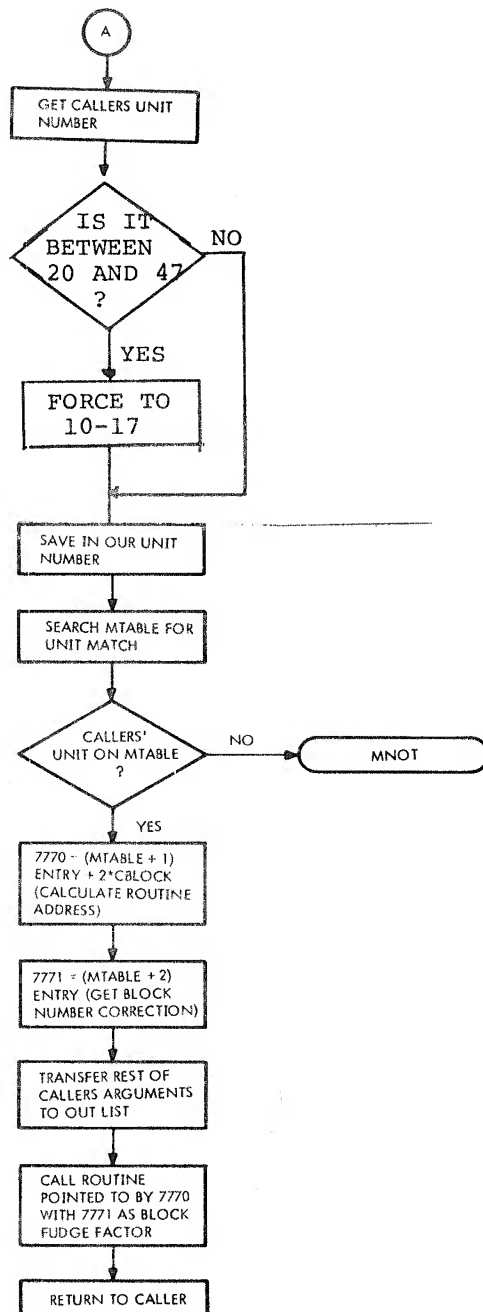




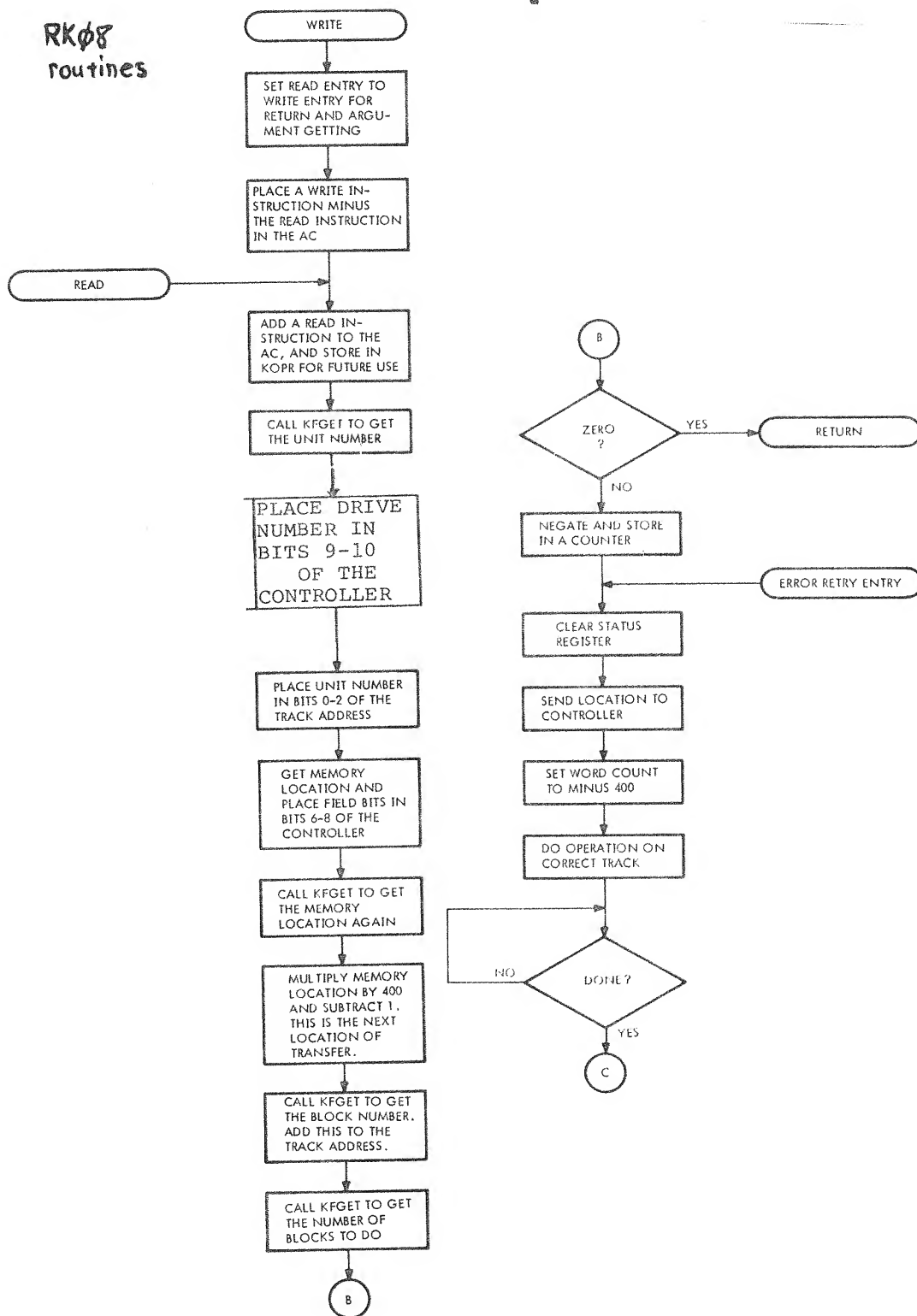


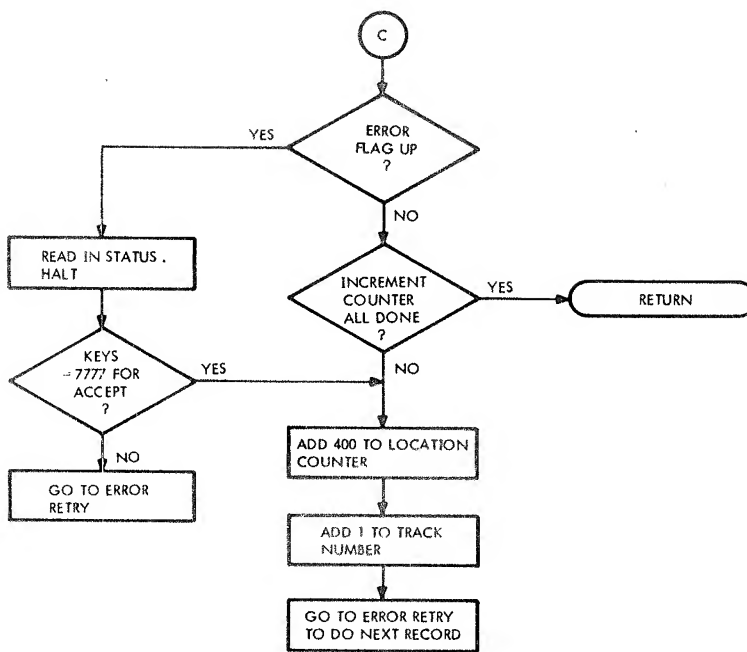
UNIT NUMBER
INITIALLY SET TO 7,
SO INCREMENTING
IT WILL PRODUCE A
10, THE FIRST DISK
UNIT



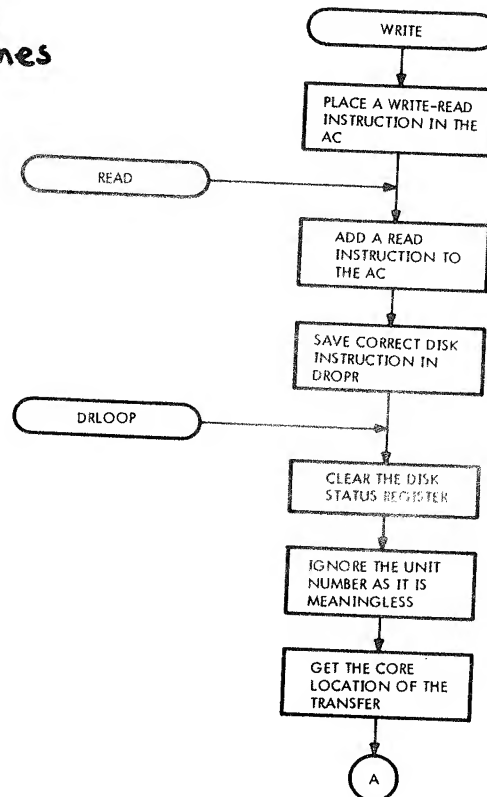


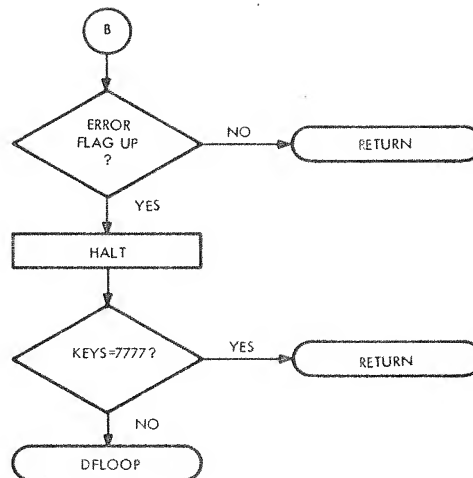
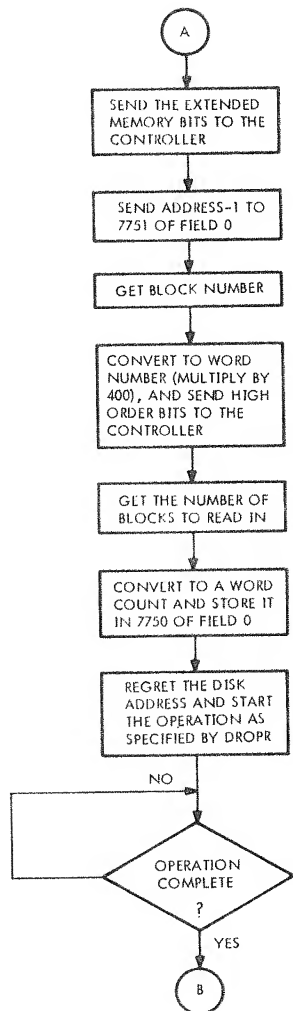
**RK08
routines**



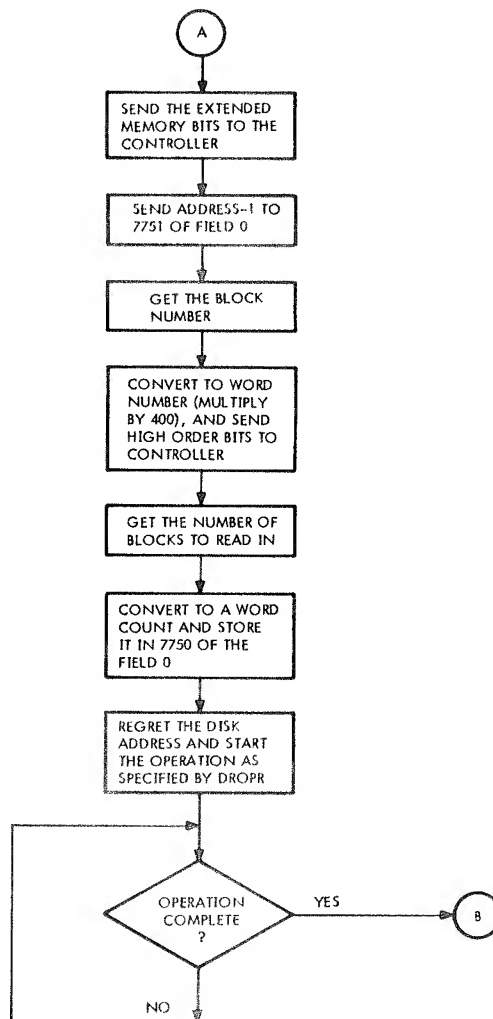
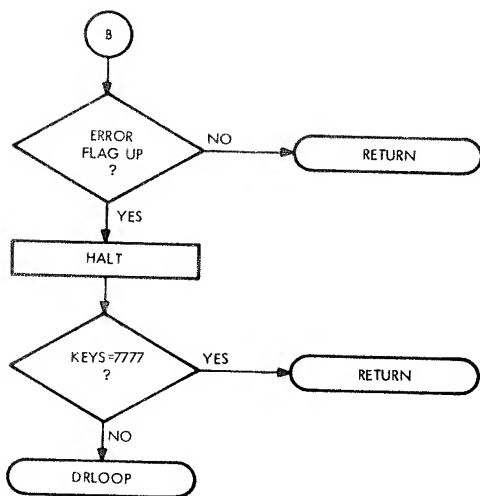
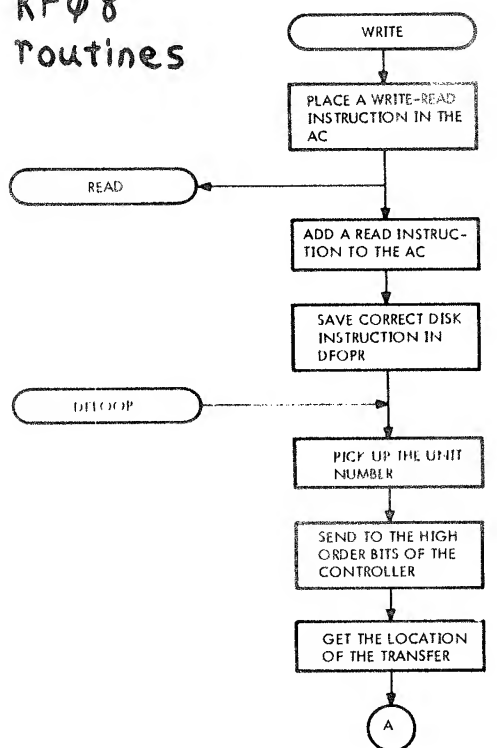


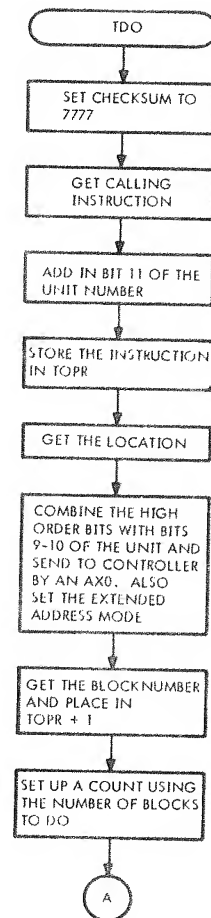
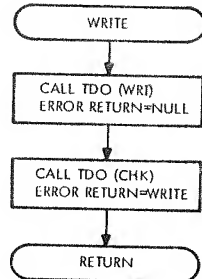
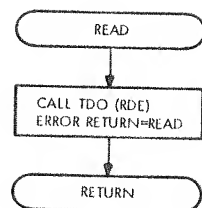
DF32 routines

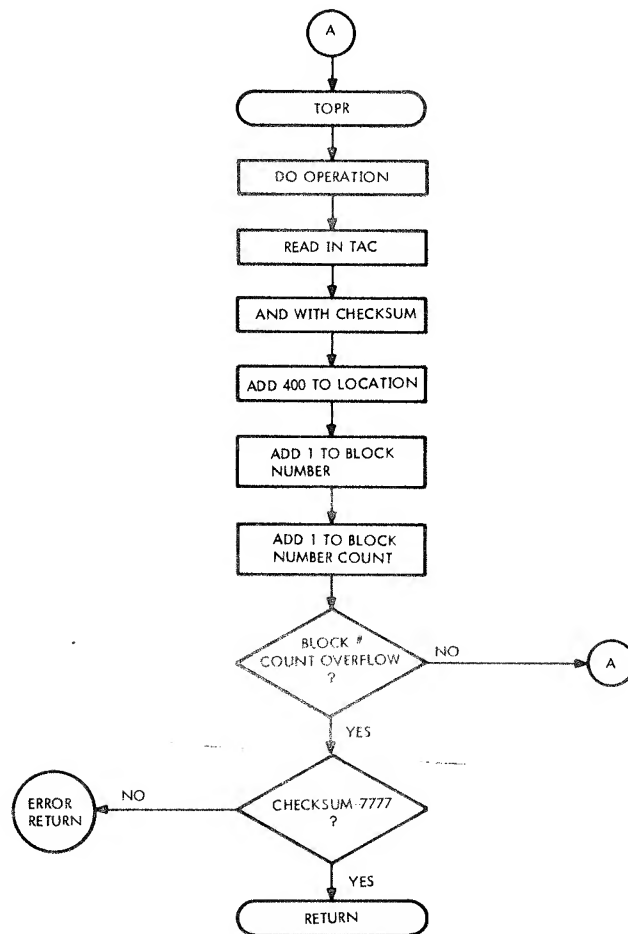




RFØ8 Routines







*20 /DIS /COF /TH

/COPYRIGHT 1970,

/THIS IS THE SYSTEM BUILD AND I-O ROUTINES ALL IN ONE.

WRITTEN BY JACK BURNES

EJECT

0023
0026
0027
0030
0031
0032
0033
0034
0035
0036
0037
0040
0041
0042
0043
0044
0045
0046
0047
0050
0051
0052
0053
0054
0055
0056
0057
0060
0061
0062
0063
0064
0065
0066
0067

PMODE

SETUP1=322
SETUP2=323
INITB=311
DI0C=345
DIHAN1=365
DIHAN2=366

/FIRST HANDLER SET UP BLOCK
/SECOND HANDLER SETUP BLOCK
/INITIALIZATION BLOCK
/CONTROLLER BLOCK
/FIRST HANDLER BLOCK
/SECOND HANDLER BLOCK

DIALUNIT=100
DIALCORE=10
DIALBLOCK=0
DIALNUMBER=10

/DEFINE THE SYSTEM UNIT NOW
/PLACE TO LOAD DIAL INTO
/PLACE WHERE DIAL RESIDES ON LOGICAL UNIT 0
/DIAL IS 8 BLOCK LONG.

EJECT

ADDRESS	INSTR	OPERAND	COMMENT
0071	FIELD	0	
0072			
0073			
0074			
0075			
0076			
0077			
0100			
0101			
0102			
0103			
0104			
0105			
0106			
0107			
0110			
0111	JMP	0020	0623
0112	READ	0021	7774
0113	WRITE	0022	7775
0114	LDA I	0023	1020
0115	20	0024	0020
0116	ESF	0025	0004
0117	LDF	0026	0647
0120	RDC	0027	0700
0121	6\DI0C	0030	6345
0122	RDC	0031	0700
0123	7\DIHAN1	0032	7365
0124	RDC	0033	0700
0125	5\DIHAN2	0034	5366
0126	CLR	0035	0011
0127	PDP	0036	0002
0130	PMODE		
0131	CLA CMA	4037	7240
0132	OCLS	4040	6742
0133	CLA	4041	7200
0134	TAD	4042	1353
0135	DLDC	4043	6732
0136	CLA	4044	7200
0137	DRDC	4045	6736
0140	TAD	4046	1361
0141	SNA CLA	4047	7650
0142	JMP I	4050	5764
0143	DCMA	4051	6601
0144	DCMA	4052	6601
0145	TAD	4053	1353
0146	DXAL	4054	6643
0147	CLA	4055	7200
0150	DXAC	4056	6645
0151	TAD	4057	1361
0152	SNA CLA	4060	7650
0153	JMP	4061	5326
0154	DCMA	4062	6601
0155	DCMA	4063	6601
0156	TAD	4064	1354
0157	DEAL	4065	6615
0160	CLA	4066	7200
0161	DEAC	4067	6616
0162	TAD	4070	1362
0163	SZA CLA	4071	7640
0164	JMP I	4072	5766
0165	TAD	4073	1373
0166	ms	4074	4313

0167	4075	4767	JMS I	AISETU
0170	4076	0000	0	
0171	4077	4770	JMS I	AISETS
0172	4100	0070	70	
0173	4101	1360	TAD	AI000
0174	4102	6615	DEAL	
0175	4103	7200	CLA	
0176	4104	6616	DEAC	
0177	4105	1363	TAD	AM1000
0200	4106	7640	SZA CLA	
0201	4107	5766	JMP I	AIDONE
0202	4110	4771	JMS I	AISETB
0203	4111	0270	270	
0204	4112	5766	JMP I	AIDONE

/CALL TO SET UP THE SOURCE WORK AREA
 /CHECK FOR TWO DISKS
 /READ BACK THE REGISTER
 /AND CHECK TO SEE IF OK
 /NOT THERE
 /SET THE BINARY NOW
 /270 IS THE CORRECT INCREMENT.
 /ALL IS WELL.

4113	0000	IMOVE,	0		/MOVES IN CORRECT CORE LOAD.
4114	6201	CFD	0		/SELECT CALLING SEQUENCE.
4115	6212	CIF	10		/ZAPPO
4116	3321	DCA	IADD		
4117	4774	JMS I	AMOVE		/DO MOVE
4120	6211	CFD	10		/SYSTEM DATA CRAP
4121	0000	0			/TO SYSTEM AREA
4122	6211	CFD	10		
4123	7430	7430			/COUNT
4124	0150	150			/RETURN TO CALLER
4125	5713	JMP I	IMOVE		

4126	1353	IRF08,	TAD	A77	/CHECK TO SEE IF FIRST DISK IS THERE
4127	4765	JMS I	AI	AICHECK	/POINTER TO THE RF08 ROUTINES.
4130	1375	TAD	ARF08		/CALL IN THE MOVERPER TO SELECT RF08.
4131	4313	JMS	IMOVE		/SET UP FOR DIAL SYSTEM
4132	4767	JMS I	AISETD		
4133	0300	300			/SET FOR SOURCE
4134	4770	JMS I	AISETS		
4135	0370	370			/SET FOR BINARY
4136	4771	JMS I	AISETB		/ALSO 370
4137	0370	370			/SET THE UNITS FOR 0 AND 1
4140	4772	JMS I	AISETU		/2 DISKS?
4141	1355	TAD	A177		
4142	4765	JMS I	AICHECK		/YEP
4143	4772	JMS I	AISETU		/3 DISKS?
4144	1356	TAD	A277		IT UP
4145	4765	JMS I	AICHECK		/JUR DISKS?
4146	4772	JMS I	AISETU		
4147	1357	TAD	A377		

Address	Instruction	Comment
0446	CIF	10
0447	JMS I	BWRITE
0450	ILOUTP	
0451	TAD	IB2
0452	TAD	LTEN
0453	DCA	IB2
0454	ISZ	BM20
0455	JMP	ILOOP
0456	/	
0457	/	
0460	IIDONE,	CIF
0461	JMS I	BWRITE
0462	DSAVI	
0463	IIDONR,	CIF
0464	JMP I	BSYSBOT
0465	/	
0466	/	
0467	/	
0470	/	
0471	/	
0472	/	
0473	ISETD,	
0474	CDF	0
0475	TAD I	ISETD
0476	ISZ	ISETD
0477	CDF	10
0500	DCA I	MSYSP2
0501	TAD	B7430
0502	DCA I	MSYSP1
0503	JMP I	ISETD
0504	/	
0505	/	
0506	/	
0507	ISETS,	
0510	CDF	0
0511	TAD I	ISETS
0512	ISZ	ISETS
0513	CDF	10
0514	DCA I	MSYSP5
0515	TAD	B7430
0516	DCA I	MSYSP4
0517	JMP I	ISETS
0520	/	
0521	/	
0522	/	
0523	ISETB,	
0524	CDF	0
0525	TAD I	ISETB
0526	ISZ	ISETB
0527	CDF	10
0530	DCA I	MSYS10
0531	TAD	B7430
0532	DCA I	MSYSP7
0533	JMP I	ISETB
0534	/	
0535	/	
0536	/	
0537	/	
0540	ISETU2,	0
0541	CDF	10
0542	ISZ	B7
0543	TAD	B7

0545	4327	2377	ISZ	BDSYS	
0546	4330	1375	TAD	B7430	
0547	4331	3777	DCA I	BDSYS	
0550	4332	2377	ISZ	BDSYS	
0551	4333	3777	DCA I	BDSYS	
0552	4334	3777	ISZ	BDSYS	
0553	4335	7240	CLA CMA		
0554	4336	3777	DCA I	BDSYS	
0555	4337	5722	JMP I	ISSETU2	
0556			/		
0557			/		
0560			/		
0561	4340	0000	ISSETU,		/BOP UP 1 UNIT
0562	4341	4322		JMS	/BOP UP TWO UNITS
0563	4342	4322		JMS	/RETURN TO CALLER
0564	4343	5740		JMP I	
0565			/		
0566			/		
0567			/		
0570			/		
0571			/		
0572			/		
0573			/		
0574			/		
0575			/		
0576			/		
0577			/		
0600			/		
0601	4344	0000	IL INP,	0	/UNIT
0602	4345	0000		0	/CORE LOC
0603	4346	0300	IB1,	300	/BLOCK NUMBER
0604	4347	0010	LTEN,	10	/NUMBER OF BLOCKS TO READ IN
0605			/		
0606			/		
0607			/		
0610			/		
0611			/		
0612	4350	0100	ILOUTP,	100	/LOGICAL UNIT 100
0613	4351	0000		0	/CORE LOCATION 0
0614	4352	0000	IB2,	0	
0615	4353	0010		10	/WRITE OUT THE 10 RECORDS,
0616			/		
0617			/		
0620			/		
0621			/		
0622			/		
0623			/		
0624			/		
0625			/		
0626	4354	0100	DSAVI,	100	/INDEX POINTER CRAP.
0627	4355	0020		20	
0630	4356	0046		46	
0631	4357	0002		2	
0632			/		
0633			/		
0634			/		
0635			/		
0636			/		
0637			/		
0641			/		
0642			/		

0644	/			
0645	/			
0646	/			
0647	/			
0650	/			
0651	/			
0652	/			
0653	/			
0654	/			
0655	/			
0656	/			
0657	/			
0660		4360	6430	BRK08, RK08
0661		4361	4113	BIMOVE, IMOVE
0662		4362	7331	MSYSP1, MSYS+1
0663		4363	7332	MSYSP2, MSYS+2
0664		4364	7334	MSYSP4, MSYS+4
0665		4365	7335	MSYSP5, MSYS+5
0666		4366	7337	MSYSP7, MSYS+7
0667		4367	7340	MSYS10, MSYS+10
0670				/
0671		4370	0150	BM7630, -7630
0672		4371	7774	BREAD, READ
0673		4372	7775	BWRITE, WRITE
0674		4373	7760	BM20, -20
0675		4374	7777	BSYSB0T, SYSB0T
0676		4375	7430	B7430, 7430
0677		4376	0007	B7, 7
0700		4377	7341	BDSYS, DSYS
0701	/			/
0702	/			/
0703	/			/
0704	/			/
0705	/			/
0706	/			/
0707	/			/
0710	/			/
0711	/			/
0712	/			/
0713	/			/
0714	/			/
0715	/			/
0716	/			/
0717	/			/
0720	/			/
0721	/			/
0722	/			/
0723	/			/
0724	/			/
0725	/			/
0726	/			/
0727	/			/
0730	/			/
0731	/			/
0732	/			/
0733	/			/
0734	/			/

EJECT

[illegible]

0767
0770
0771
0772
0773
0774
0775
0776
0777
1000
1001
1002
1003
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*6430

RK08=,

/THE RK08 ROUTINES GO HERE,

*6430

/ALL ROUTINES ARE ORIGINATED AT 30 OF THE PAGE,

DCLA=6751
DLDC=6732
DLDR=6733
DLDW=6735
DCHP=6737
DRDA=6734
DRDC=6736
DRDS=6741
DCLS=6742
DMNT=6743
DSKC=6745
DSKT=6746
DSKE=6747
DRWC=6752
DLWC=6753
DLCA=6755
DRCA=6757

[illegible]

1102	0211	0142	DSKC	/WAIT FOR COMPLETION NOW,	
1166	6520	7410	SKP	/SKIP IF NOT DONE	
1167	6521	5345	JMP	/IF TEST WHETHER MORE RECORDS TO GO	
1170	6522	6747	DSKE	/IS THERE AN ERROR FLAG UP???	
1171	6523	5317	JMP	/NOPE, WAIT FOR EITHER COMPLETION OR ERROR FLAG,	
1172					
1173	6524	6741	DRDS	/READ IN THE STATUS WORD IF ERROR OCCURED,	
1174	6525	7402	HLT	/BRING CPU TO A GRINDING HALT,	
1175	6526	7604	LAS	/READ IN THE KEYS NOW,	
1176	6527	3232	DCA	/SAVE THE KEYS FOR A SECOND	
1177	6530	7240	CLA CMA	/NOW PREPARE TO CLEAR THE STATUS REGISTER	
1200	6531	6742	DCLS	/CLEAR THE STATUS REGISTER	
1201	6532	7200	CLA	/CLEAR THE AC TO BE SURE	
1202	6533	6751	DCLA	/ZAP THE DISK OVER, JUST IN CASE,	
1203	6534	6747	DSKE	/IS THE ERROR FLAG UP AGAIN?	
1204	6535	7410	SKP	/NOPE, TEST COMPLETION FLAG,	
1205	6536	5324	JMP	/ERROR FLAG UP, GIVE ERROR AGAIN,	
1206	6537	6745	DSKC	/COMPLETION FLAG UP,	
1207	6540	5334	JMP	/NOPE, WAIT AROUND,	
1210	6541	1232	TAD	/NOW RETRIEVE THE KEYS,	
1211	6542	7040	CMA	/COMPLEMENT THE KEYS NOW,	
1212	6543	7640	SZA CLA	/ACCEPT???	
1213	6544	5306	JMP	/NOPE, RETRY THIS BLOCK NOW,	
1214					
1215					
1216					
1217					
1220					
1221	6545	2350	KEND,	/BOP UP THE COUNTER,	
1222	6546	5302	ISZ	/NOT DONE, GET THE NEXT RECORD,	
1223	6547	5630	JMP I	/FINISHED, RETURN TO THE CALLER	
1224					
1225					
1226					
1227					
1228					
1229					
1230					
1231					
1232					
1233					
1234					
1235	6550	0000	KCOUNT, 0	/CONTAINS -NUMBER OF RECORDS TOGO	
1236	6551	0000	KTRACK, 0	/CONTAINS CURRENT TRACK AND SECTOR,	
1237	6552	0000	KLOC, 0	/CONTAINS CURRENT LOC-1 BEING USED,	
1240					
1241					
1242					
1243					
1244					
1245	6553	0000	KFGET, 0		
1246	6554	1630	TAD I	/GET THE ARGUMENT NOW	
1247	6555	2230	ISZ	/PUSH TO NEXT ARGUMENT	
1250	6556	5753	JMP I	/RETURN TO THE CALLER,	
1251					
1252					
1253					
1254					
1255					
1256					
1257	6557	0007	KL 7,		
1260	6560	0030	KL 30,		
1261	6561	0070	KL 70,		
1262	6562	0400	KL 400,		
1263					

JMS
misc line 8
Return to -64

1264	6564	7777	KL7777, 7777
1265			/
1266	6565	6733	KDLOR, DLOR
1267			/
1270			/
1271			/
1272			/
1273			/
1274			/
1275			/
1276			/
1277			/
1300			/
1301			/
1302			/
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1313			/
			EJECT

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*6630

DF32=.

*6630

6630	0000	DRREAD, 0	JMP	DR1	/DF32 READ-WRITE ROUTINES,
6631	5236	DRWRIT, 0	TAD	DRWRIT	/INTO MAIN PROCESSOR
6632	0000	DCA	CLA CLL	DRREAD	/WRITE ENTRY POINT
6633	1232	TAD	DRMAR	CML RTL	/SET UP ARG GETTER
6634	3230	CLA CLL	DROPR	DRMAR	/+2
6635	7326	TAD	DCA	DROPR	/ADD IN READ INSTRUCTION,
6636	1327	DCA	DCMA	DRREAD	/SAVE THE CORRECT INSTRUCTION,
6637	3303	DCMA	ISZ	DRREAD	/CLEAR THE STATUS REGISTER
6640	6601	TAD	RAR	DRREAD	/UNT MEANINGLESS FOR DF32S
6641	2230	AND	AND	DRL70	/GET THE CORE LOCATION
6642	1630	TAD I	DCA	DRWRIT	/SHIFT OVER 1 PLACE
6643	7010	RAR	RIF	DRWRIT	/ZAP OUT CRAP
6644	0330	AND	TAD	DRCDF	/AND SAVE AWAY FOR NOW,
6645	3232	DCA	DCA	DRXCOF	/GET THIS FIELD
6646	6224	RIF	TAD	DRREAD	/ADD IN A CDF 0
6647	1253	TAD	DCA	DRRTR	/NOW STORE FIELD RESTORE INSTRUCTION,
6650	3333	DCA	TAD I	0	/REGET THE CORE LOCATION
6651	1630	TAD I	JMS		/ROR 5 AND AND WITH 7400,
6652	4335	JMS	CDF		/SET TO SFTUP DATA BREAK LOCATIONS
6653	6201	CDF			
6654	7010				

1413	6655	7040	CMA	I		/SUBTRACT 1 FOR A GOOD LOCATION
1414	6656	3745	DCA	I	DRMAD	/STORE AWAY FOR NOW
1415	6657	4332	JMS		DRXX	/RESET TO THIS DATA FIELD
1416	6660	2230	ISZ		DRREAD	/UP UP TO BLOCK NUMBER
1417	6661	1630	TAD	I	DRREAD	/GET THE BLOCK NUMBER
1420	6662	7006	RTL		ORL3700	/ROTATE TO BITS 1-5 OF THE AC
1421	6663	0346	AND		DRWRIT	/AND OUT CRAP
1422	6664	1232	TAD			/ADD IN MEMORY FIELD
1423	6665	6615	DEAL			/SEND TO CONTROLLER,
1424	6666	7200	CLA			
1425	6667	1630	TAD	I	DRREAD	/GET THE BLOCK NUMBER AGAIN
1426	6670	4335	JMS		DRTR	/INTO 7400S BITS
1427	6671	3232	DCA		DRWRIT	/AND SAVE FOR THE DROP
1430	6672	2230	ISZ		DRREAD	/BOP TO NUMBER OF RECS,
1431	6673	1630	TAD	I	DRREAD	/GET THE NUMBER OF RECS,
1432	6674	2230	ISZ		DRREAD	/BOP PAST THE CRAP FOR RETRUN
1433	6675	4335	JMS		DRTR	/ROTATE INTO 7400S BITS
1434	6676	6201	COF		Ø	/SET TO Ø FOR DATA BREAK
1435	6677	7041	CIA			/NEGATE
1436	6700	3744	DCA	I	DRMC	/DATA BREAK SET UP NOW
1437	6701	4332	JMS		DRXX	/RESTORE THIS FIELD
1440	6702	1232	TAD		DRWRIT	/GET LOW ORDER BITS OF BLOCK NUMBER
1441	6703	0000	Ø			/XEC. DISK INSTRUCTION
1442	6704	7200	CLA			
1443	6705	6621	DFSE			
1444	6706	5314	JMP		DRRR	/WAIT FOR DONE OR ERROR FLAG,
1445	6707	6622	DFSC			
1446	6710	5305	JMP		-3	/HANG AROUND FOR A WHILE,
1447	6711	6621	DFSE			/CHECK ERROR FLAG TO BE SURE
1450	6712	5314	JMP		DRRR	/ERROR!
1451	6713	5352	JMP		DRCLRF	/EXIT, ALL IS WELL
1452			/			
1453			/			
1454	6714	6616	DEAC			/READ IN THE ERROR REG.
1455	6715	7402	HLT			/HALT NOW.
1456	6716	7604	LAS			/GET HIS REQUEST.
1457	6717	7040	CMA			/NEGATE FOR TEST
1460	6720	7650	SNA	CLA	DRCLRF	/EXIT NOW
1461	6721	5352	JMP		CLA CLL IAC RTL	/+4
1462	6722	7307	CLA	CLL		/-4
1463	6723	7041	CIA			/RESET ARG POINTER
1464	6724	1230	TAD		DRREAD	/NOW
1465	6725	3230	DCA		DRREAD	/AND TRY AGAIN,
1466	6726	5240	JMP		DRLOOP	
1467			/			
1470			/			
1471			/			
1472			/			
1473	6727	6603	DMAR,			
1474	6730	0070	ORL70,	70		
1475	6731	0300	DRL300,	300		
1476	6732	0000	DRXX,	Ø		
1477	6733	0000	DRXCDF,	Ø		
1500	6734	5732	JMP	I	DRXX	
1501			/			
1502	6735	0000	DRRTR,			
1503	6736	7012	RTR			
1504	6737	7012	RTR			
1505	5740	7010	RAR			
1506	6741	0343	AND		DRL7400	
1507	6742	5735	JMP	I	DRRTR	
1510			/			

1512	6744	DRMNC,	7750				
1513	6745	DRMAD,	7751				
1514	6746	DRL3700,	3700				
1515		/					
1516		/					
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1524							
1525							
1526							
1527							
1530		DRCLRF,	DCMA				
1531	6752		6601				
1532	6753		6601				
1533	6754	JMP I	DRREAD				
1534							
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/ORIGIN PASS DATA BREAK LOCATIONS,

*,87600+152

/CLEAR DF32 FLAGS IN CASE HE'S USING INTERRUPTS.
/YOU REALLY NEED TWO OF THEM.
/AND RETURN,

EJECT

Address	Instruction	Comment
1560	/	
1561	/	
1562	/	
1563	/	
1564	/	
1565	/	
1566	/	
1567	/	
1570	/	
1571	CLA	/CLEAR THE AC TO BE SURE
1572	IOF	/TURN OFF INTERRUPTS IN CASE WE'RE IN FOCAL 12.
1573	RDF	/GET THE CALLING FIELD
1574	TAD	/ADD IN THE BASE CDF
1575	DCA	/AND SAVE AWAY
1576	RIF	/READ IN THIS FIELD
1577	TAD	/AND SET UP THIS DATA FIELD CHANGE
1600	DCA	/STASH AWAY.
1601	DCA	/CLEAR THE REQUEST WORD
1602	STCDF	/SET DATA FIELD TO THIS FIELD
1603	TAD I	/GET THE READ ADDRESS
1604	SNA	/WAS IT A READ?
1605	ISZ	/NOPE. A WRITE. SET UP CALLING ADDRESS
1606	TAD I	/ADD IN WRITE CALL LOCATION.
1607	DCA	/AND STASH AWAY.
1610	DCA I	/CLEAR THE READ LOCATION
1611	DCA I	/AND THE WRITE LOCATION FOR NEXT TIME THROUGH
1612	TAD	/GET THE POINTER TO THE UNITS TABLE.
1613	DCA	/AND SAVE AWAY IN A TEMPORARY.
1614	JMS	/SET TO CALLERS FIELD
1615	TAD I	/GET THE START OF CALLERS ARGUMENTS.
1616	DCA	/STASH AWAY
1617	ISZ	/AND BOP UP FOR THE RETURN
1620	TAD I	/GET THE USERS UNIT NOW.
1621	TAD	/IS IT GREATER OR EQUAL TO 20?
1622	SPA	/NOPE. ITS LESS THEN 20.
1623	JMP	/IS IT GREATER OR EQUAL TO 50?
1624	TAD	
1625	SMA	/BIGGER THEN 47.
1626	JMP	/ITS BETWEEN 20 AND 47. AN EXTENDED UNIT. GET
1627	AND	/THE UNITS BITS AND MAKE INTO 1X TYPE NUMBER
1630	TAD	/MAKE BACK INTO ORIGINAL NUMBER NOW.
1631	TAD	/DEPENDING ON WHERE RANGE OCCURED.
1632	LESS20,	/AND STASH AWAY NOW FOR THE TEST.
1633	DCA	/SET TO THIS CDF
1634	JMS	
1635	/	
1636	/	
1637	/	
1640	ML00P,	/GET NEXT UNIT ON TABLE
1641	SPA	/MINUS=END OF LIST???
1642	JMP	/UNIT NOT THERE
1643	CIA	
1644	TAD	/GET THE CALLING UNIT
1645	7050	/A MATCH???
1646	7051	/SURE IS
1647	7052	/+3
1648	CLA:CLL	/BOP TO NEXT ENTRY ON TABLE
1649	7053	/AND PLACE BACK
1650	7054	
1651	7055	
1652	7056	/TRY THIS ONE.

15 16
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 19 20
 21 22

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/GET OVER INTO SUPER MODE

/SET FOR FULL SIZE LETTERS
/ZAP IT OUT NOW
/NOW DISPLAY IT, BABY,

MNL

LINC
LMODE
LDA I
200
ESF
JMP

PMODE

EJECT

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Line	Address	Instruction	Comment
1726	7064	ISZ TAD	/PUSH PAST TO NEXT ITEM
1727	7065	CLL RAL	/GET WHETHER READ OR WRITE(0 OR 1)
1728	7066	TAD I	/MAKE IT A TWO
1729	7067	DCA I	/GET THE CORRECT ROUTINE ADDRESS
1730	7070	ISZ TAD	/STORE IN 7770
1731	7071	ISZ TAD	/PUSH PAST THE CRAP
1732	7072	ISZ TAD	/GET THE FUDGE FACTOR
1733	7073	ISZ TAD	/AND STORE AWAY
1734	7074	ISZ TAD	/REGET THE ROUTINE ADDRESS
1735	7075	ISZ TAD	/SAVE IN TEMP FOR NOW,
1736	7076	ISZ TAD	/SET TO CALLERS CDF
1737	7077	ISZ TAD	/REGET THE USERS UNIT NOW,
1738	7100	ISZ TAD	/AND STASH AWAY,
1739	7101	ISZ TAD	/PUSH TO CORE LOC
1740	7102	ISZ TAD	/GET THE CORE LOC
1741	7103	ISZ TAD	/AND STASH AWAY
1742	7104	ISZ TAD	/GET TO BLOCK NUMBER
1743	7105	ISZ TAD	/SET TO THIS DATA FIELD
1744	7106	ISZ TAD	/RECALL THE FUDGE FACTOR
1745	7107	ISZ TAD	/SET TO CALLERS LOC
1746	7110	ISZ TAD	/ADD IN USERS BLOCK NUMBER
1747	7111	ISZ TAD	/NOW SAVE AWAY
1748	7112	ISZ TAD	/BOP TO NEXT ONE
1749	7113	ISZ TAD	/SAVE THE NUMBER OF BLOCKS NOW,
1750	7114	ISZ TAD	/SET TO THIS CDF
1751	7115	ISZ TAD	/CALL THE CORRECT ROUTINE NOW,
1752	7116	ISZ TAD	
1753	7117	ISZ TAD	
1754	7120	ISZ TAD	
1755	7121	ISZ TAD	
1756	7122	ISZ TAD	
1757	7123	ISZ TAD	
1758	7124	ISZ TAD	
1759	7125	ISZ TAD	
1760	7126	ISZ TAD	
1761	7127	ISZ TAD	
1762	7128	ISZ TAD	
1763	7129	ISZ TAD	
1764	7130	ISZ TAD	
1765	7131	ISZ TAD	
1766	7132	ISZ TAD	
1767	7133	ISZ TAD	
1768	7134	ISZ TAD	
1769	7135	ISZ TAD	
1770	7136	ISZ TAD	
1771	7137	ISZ TAD	
1772	7138	ISZ TAD	
1773	7139	ISZ TAD	
1774	7140	ISZ TAD	
1775	7141	ISZ TAD	
1776	7142	ISZ TAD	
1777	7143	ISZ TAD	
1778	7144	ISZ TAD	
1779	7145	ISZ TAD	
1780	7146	ISZ TAD	
1781	7147	ISZ TAD	
1782	7148	ISZ TAD	
1783	7149	ISZ TAD	
1784	7150	ISZ TAD	
1785	7151	ISZ TAD	
1786	7152	ISZ TAD	
1787	7153	ISZ TAD	
1788	7154	ISZ TAD	
1789	7155	ISZ TAD	
1790	7156	ISZ TAD	
1791	7157	ISZ TAD	
1792	7158	ISZ TAD	
1793	7159	ISZ TAD	
1794	7160	ISZ TAD	
1795	7161	ISZ TAD	
1796	7162	ISZ TAD	
1797	7163	ISZ TAD	
1798	7164	ISZ TAD	
1799	7165	ISZ TAD	
1800	7166	ISZ TAD	
1801	7167	ISZ TAD	
1802	7168	ISZ TAD	
1803	7169	ISZ TAD	
1804	7170	ISZ TAD	
1805	7171	ISZ TAD	
1806	7172	ISZ TAD	
1807	7173	ISZ TAD	
1808	7174	ISZ TAD	
1809	7175	ISZ TAD	
1810	7176	ISZ TAD	
1811	7177	ISZ TAD	
1812	7178	ISZ TAD	
1813	7179	ISZ TAD	
1814	7180	ISZ TAD	
1815	7181	ISZ TAD	
1816	7182	ISZ TAD	
1817	7183	ISZ TAD	
1818	7184	ISZ TAD	
1819	7185	ISZ TAD	
1820	7186	ISZ TAD	
1821	7187	ISZ TAD	
1822	7188	ISZ TAD	
1823	7189	ISZ TAD	

2026	7136	6201	LCDF,	COF	
2027	7137	7774	MREAD,	7774	
2030	7140	7775	MWRITE,	7775	
2031	7141	7300	PTABLE,	MYTABLE	
2032	7142	0000	TEMP,	0	
2033	7143	0000	MRETL,	0	
2034	7144	0000	MCALL,	0	
2035	7145	0006	L6,	6	
2036	7146	0007	L7,	7	
2037	7147	0020	L20,	20	
2040	7150	0030	L30,	30	
2041	7151	7770	L7770,	7770	
2042	7152	7771	L7771,	7771	
2043	7153	7760	M20,	-20	
2044	7154	7750	M30,	-30	
2045	7155	7740	M40,	-40	
2046					
2047					
2050					
2051					
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2061					
2062					
2063					
2064	1156	0002	BOOTER,	PDP	
2065				PMODE	
2066	7157	0345	AND	L6	
2067	7160	7650	SNA CLA		
2070	7161	5365	JMP	RK080K	
2071	7162	6751	DCLA		
2072	7163	6745	DSKC		
2073	7164	5363	JMP	"1	
2074	7165	3740	RK080K,	DCA I	
2075	7166	4737		MWRITE	
2076	7167	7174		JMS I	
2077	7170	6141		MREAD	
2100				DPBOOT	
2101	1171	0643		LINC	
2102	1172	0602		PMODE	
2103	1173	6021		LDF	
2104				LIF	
2105				JMP	4021
2106					
2107					
2110					
2111					
2112					
2113	7174	0100	DPBOOT,	DIALUNIT	
2114	7175	0010		DIALCORE	
2115	7176	0000		DIALBLOCK	
2116	7177	0010		DIALNUMBER	
2117					
2120					
2121					
2122					
2123					

/TELL ASSY WE ARE IN LINC MODE WHEN WE GET HERE
 /GET OVER INTO THE GOOD MODE,
 /SEE WHAT THE AC CONTAINS,
 /BITS 9 OR 10 ON?
 /NOPE, DISK IS OK, ALL IS WELL,
 /DISK WAS ON WRONG DRIVE, REPOSITION NOW,
 /AND WAIT FOR IT,BABY
 /HO HUM
 /ZAP OUT THE WRITE ENTRY IN CASE ROUTINES BOMBED,
 /CALL THE READ ROUTINE
 /POINTER TO BOOTSTRAP INFO
 /GET OVER INTO LINC MODE
 /SET TO DO STANDARD CALL
 /JMP TO SYSTEM RESTART LOCATION,
 /GET BACK INTO 8 MODE

1533

Address	Instruction	Comment
2145	/	
2146	/	
2147	/	
2150	/	
2151		*7200
2152	/	
2153	/	
2154	/	
2155	/	
2156	/	
2157	MOVE,	
2160	0000	
2161	7200	
2162	7202	CLA
2163	4236	JMS
2164	3222	DCA
2165	4236	JMS
2166	3244	DCA
2167	4236	JMS
2168	3224	DCA
2169	4236	JMS
2170	3245	DCA
2171	6214	RDF
2172	1246	TAD
2173	3234	DCA
2174	4236	JMS
2175	7450	SNA
2176	5234	JMP
2177	7041	CIA
2200	3236	DCA
2201	0000	SWIN,
2202	1644	TAD I
2203	0000	SWOUT,
2204	3645	DCA I
2205	2245	ISZ
2206	7000	NOP
2207	2244	ISZ
2210	7000	NOP
2211	2236	ISZ
2212	5222	JMP
2213	0000	SWRET,
2214	5600	JMP I
2215		
2216		
2217		
2220		
2221	0000	SWGET,
2222	1600	TAD I
2223	7240	ISZ
2224	5636	JMP I
2225		
2226		
2227		
2230	0000	SWTMP1,
2231	0000	SWTMP2,
2232	0000	SWLIN,
2233	0000	SWLOUT,
2234	6203	SLCDF,
2235		
2236		
2237		
2240		
2241		
2242		
7247	3077	DTABLE, 3077
7248		

7251	0000		0000	/THE SPACE BETWEEN THE N AND THE O
7252	4177		4177	/END OF THE "NO" MESSAGE
7253	7741		7741	
2244		/		
2245		/		
2246		/		
2247		/		
2250		/		
2251		/		
2252		/		
2253		/		
2254		/		
2255		/		
2256		/		
2257		/		
2260		/		
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2262		/		
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2277		/		
2300		/		
2301		/		
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2336		/		
7		/		
0		/		
2341		/		

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*7300

/THIS IS WHERE THE UNIT TABLE RESIDES.

7300 0000 MTABLE, 0
7301 7630 7630
7302 0000 0
7303 0001 1
7304 7630 7630
7305 0000 0
7306 0002 2
7307 7630 7630
7310 0000 0
7311 0003 3
7312 7630 7630
7313 0000 0
7314 0004 4
7315 7630 7630
7316 0000 0
7317 0005 5
7320 7630 7630
7321 0000 0
7322 0006 6
7323 7630 7630
7324 0000 0
7325 0007 7
7326 7630 7630
7327 0000 0
7330 0100 100
7331 7630 7630
7332 0300 300
7333 0110 110
7334 7630 7630
7335 0370 370
7336 0111 111
7337 7630 7630
7340 0370 370
7341 7777 7777

MSYS,

DSYS,

/START OF DISK UNITS OR END OF TABLE

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EJECT

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Address	Instruction	Comment
2464		
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2500		
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2561		

/AND SAVE FOR A SECOND,
/AND NOW GO DO THE REBOOTSTRAP,

DX61777
CHECKF

STC
JMP

PMODE

EJECT

1426 4801
1427 7254

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DISK DEFINITIONS (DF32-RF08)

DCMA=6601
DMAR=6603
DMAW=6605
DCEA=6611
DSAC=6612
DEAL=6615
DEAC=6616
DFSE=6621
DISK=6623
DIML=6615
DFSC=6622
DIWA=6616
DMAC=6626
DMMT=6646
DCXA=6641
DXAL=6643
DXAC=6645

*7430

RF08=.

/DEFINE THE RF08 LOCATIONS.

47430

100 S
DCHECK
where
FEDRUS 12

Address	Instruction	Comment
3025	CDP	
3026	TAD I	DFMWC
3027	JMS	DFXX
3030	SNA CLA	
3031	JMP I	DFREAD
3032	DFBAD1,	DIMA
3033	HLT	
3034	LAS	
3035	CMA	
3036	SNA CLA	
3037	JMP I	DFREAD
3040	CLA CLL	IAC RTL
3041	CIA	
3042	TAD	DFREAD
3043	DCA	DFREAD
3044	JMP	DFLOOP
3045		
3046		
3047		
3050		
3051		
3052		
3053		
3054		
3055		
3056	DMAR	
3057	DFMAR,	
3060	DFGET,	
3061	TAD I	DFREAD
3062	ISZ	DFREAD
3063	JMP I	DFGET
3064	7751	
3065		
3066		
3067		
3070		
3071		
3072		
3073		
3074		
3075	DFRTR,	
3076		
3077	RTR	
3100	RAR	
3101	AND	
3102	JMP I	DFL7400
3103	70	DFRTR
3104	DFXX,	
3105	DFXCDF,	
3106	JMP I	DFXX
3107	DFL7400,	7400
3110	DFL377,	377
3111	DFL1003,	1003
3112	DFMWC,	7750
3113		
3114		
3115		
3116		
3117		
3120		
3121		
3122		

3124 /
3125 /
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3140 /
3141 /
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3163 /
3164 /
3165 /
3166 /
3167 /

7570 0000
7571 0000
7572 7000
7573 7422
7574 0000
7575 0000
7576 5772
7577 5773

*7570

0
0
MAIN
BOOTCK
0
0
JMP I :-4
JMP I :-4

EJECT

[illegible]

3267
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EJECT

Address	Instruction	Comment
3277	/	
3300	/	
3301	/	
3302	/	
3303	/	
3304	/	
3305	/	
3306	/	
3307	/	
3310	/	
3311	/	
3312	/	
3313	/	
3314	/	
3315	/	
3316	TREAD, 0	7630 0000
3317	JMP	7631 5247
3320	TWRITE, 0	7632 0000
3321	TAD	7633 1232
3322	DCA	7634 3230
3323	JMS	7635 4253
3324	LMODE	
3325	WRI	1636 0706
3326	PMODE	
3327	TL400,	7637 0400
3330		7640 4253
3331	JMS	
3332	CHK	1641 0707
3333	PMODE	
3334	JMP	7642 5233
3335	/	
3336	TTRET,	7643 7307
3337	TAD	7644 1230
3340	DCA	7645 3230
3341	JMP I	7646 5630
3342	/	
3343	TTREAD,	7647 4253
3344	LMODE	
3345	ROE	1650 0702
3346	PMODE	
3347	JMP	7651 5247
3350	JMP	7652 5243
3351	/	
3352	/	
3353	/	
3354	/	
3355	/	
3356	TDO:	7653 0000
3357	TAD	7654 1230
3360	DCA	7655 3350
3361	TAD I	7656 1653
3362	ISZ	7657 2253
3363	DCA	7660 3331
3364	CLA CMA	7661 7240
3365	DCA	7662 3362
3366	TAD I	7663 1750
3367	AND	7664 0356
3370	CLL RAR	7665 7110
3371	DCA	7666 3327
3372	RTL	7667 7006
3373	RTL	7670 7006
3374	TAD	7671 1331

3377	7674	1750	TAD I	TCALLY	/GET THE CORE LOCATION NOW,
3400	7675	0360	AND	TL160	/JUST GET THE FIELD BITS NOW,
3401	7676	7106	CLL RTL		/AND ROTATE TO BITS
3402	7677	7006	RTL		/0-2
3403	7700	7004	RAL		
3404	7701	1327	TAD	TAXO	/ADD IN THE TAPE REGISTER WORD,
3405	7702	1357	TAD	TL20	/ALSO ADD IN THE EXTENDED ADDRESS BIT,
3406	7703	3327	OCA	TAXO	/AND SAVE IT AWAY,
3407	7704	1750	TAD I	TCALLY	/GET THE CORE LOCATION AGAIN,
3410	7705	7012	RTR		/ROTATE TO BITS 0-3,
3411	7706	7012	RTR		
3412	7707	7010	RAR		
3413	7710	0361	AND	TL7400	/AND AND OUT STRAY CRAP,
3414	7711	3232	OCA	TWRITE	/AND SAVE AWAY, TWRITE IS FREE NOW.
3415	7712	2350	ISZ	TCALLY	/BOP TO THE BLOCK NUMBER
3416	7713	1750	TAD I	TCALLY	/GET THE BLOCK NUMBER
3417	7714	3332	DCA	TBLOCK	/AND SAVE AWAY,
3420	7715	2350	ISZ	TCALLY	/BOP TO THE NUMBER OF BLOCKS TO READ IN,
3421	7716	1750	TAD I	TCALLY	/GET THE NUMBER NOW,
3422	7717	7041	CIA		/NEGATE,
3423	7720	7450	SNA	TOUT	/NO BLOCKS?
3424	7721	5352	JMP	TCOUNT	/NO BLOCKS, RETURN,
3425	7722	3351	DCA		/SET UP THE COUNT NOW,
3426					
3427					
3430	7723	1232	TAD	TWRITE	/GET THE CORE ADDRESS,
3431	7724	6141	LINC		/GET INTO LINC MODE
3432			LMODE		
3433	1725	0023	TMA		/SEND TO THE CONTROLLER,
3434	1726	1020	LDA I		/GET THE FUNCTIONS,
3435	1727	0000	0		/AND SEND IT TO THE CONTROLLER,
3436	1730	0001	AXO		/WITH AN AXO,
3437					
3440	1731	0000	0		/NOW DO THE ACTUAL TAPE INSTRUCTION,
3441	1732	0000	0		/WITH THE BLOCK NUMBER HERE, THIS HANGS UNTIL DONE,
3442					
3443	1733	0002	PDP		/BACK INTO 8 MODE
3444			PHODE		/FOR NORMAL CRAP
3445					
3446	7734	0362	AND	TRET	/CHECK FOR ALL 7775
3447	7735	3362	OCA	TRET	
3450	7736	1237	TAD	TL400	/ADD 400 TO THE LOC
3451	7737	1232	TAD	TWRITE	
3452	7740	3232	DCA	TWRITE	
3453	7741	2332	ISZ	TBLOCK	/BOP UP THE BLOCK NUMBER
3454	7742	2351	ISZ	TCOUNT	/ALL DONE???
3455	7743	5323	JMP	TLOOP	/NOPE, DO SOMMSE MORE.,,
3456	7744	5352	JMP	TOUT	/ALL DONE, CHECK CHECKSUM AND EXIT,
3457					
3460					
3461					
3462					
3463					
3464	7750	0000	*7750		
3465	7751	0000	TCALLY, 0		/PLACE OF D.B. LOCATIONS,
3466			TCOUNT, 0		
3467					
3470					
3471					
3472	7752	2362	TOUT,	TRET	THE CHECKSUM OK???
3473	7753	5653	ISZ	TDO	/NOPE, EXIT
3474			JMP I		
3475					

	JMP I	T00	/RETURN,
3476	/		
3477	/		
3500	/		
3501	/		
3502	/		
3503	/		
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3552	/		
3553	/		

7756	0007	TL7,	7
7757	0020	TL20,	20
7760	0160	TL160,	160
7761	7400	TL7400,	7400
7762	0000	TRET,	0
*7770			
7770	0000		
7771	0000		
7772	7000	MAIN	
7773	7422	BOOTCK	
7774	0000	READ,	
7775	0000	WRITE,	
7776	5772	SYSBOT,	JMP I 7772
7777	5773		JMP I 7773
EJECT			

/SYSTEM COMMUNICATIONS REGION

/CONTAINS THE ACTIVE ADDRESS OF I/O

/CONTAINS THE NECESSARY FUDGE FACTOR

/POINTER TO I/O INTERPRETER

/POINTS TO BOOTSTRAP

/READ ENTRY POINT

/WRITE ENTRY POINT

/DOES THE CORRECT OPERATION

/CALLS THE SYSTEM BOOTSTRAP,

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SYMBOL	VALUE	DEF	REFERENCES
ADF32	4173	0330	0165
AICHEC	4165	0322	0244 0257 0262 0265
AIDONE	4166	0323	0164 0201 0204 0267
AIRK08	4164	0321	0142
A1SET8	4171	0326	0202 0253
A1SETD	4167	0324	0167 0247
A1SETS	4170	0325	0171 0251
A1SETU	4172	0327	0255 0260 0263 0266
AMOVE	4174	0331	0223
AM100	4162	0315	0162
AM1000	4163	0316	0177
AM77	4161	0314	0140 0151
ARF08	4175	0332	0245
A100	4154	0307	0156
A1000	4160	0313	0173
A177	4155	0310	0256
A277	4156	0311	0261
A377	4157	0312	0264
A77	4153	0306	0134
BDSYS	4377	0700	0544 0545 0547 0550 0551 0552 0554
B1MOVE	4361	0661	0371
BM20	4373	0674	0454
BM7630	4370	0671	0431
BOOTCK	7422	2555	3135 3535
BOOTER	7156	2064	2317
BREAD	4371	0672	0436 0441
BRK08	4360	0660	0370
BSYSB0	4374	0675	0464
BWRITE	4372	0673	0447 0461
B7	4376	0677	0542 0543
B7430	4375	0676	0501 0515 0531 0546
CBLOCK	7121	1771	1601 1605 1735 1761
CCORE	7120	1770	1753
CHB00T	7272	2314	2305
CHECKF	7254	2272	2564
CNUM	7122	1772	1764
CUNIT	7117	1767	1633 1644 1750
DCEA	6611	2645	2527 2532 2533 2534 2535 2536
DCHAR	6002	0754	
DCHP	6737	1036	
DCLA	6751	1032	1202 2071
DCLS	6742	1042	0132 1153 1200
DCMA	6601	2642	0143 0144 0154 0155 1376 1530 1531 2744
DCXA	6641	2660	
DEAC	6616	2650	0161 0176 1454
DEAL	6615	2647	0157 0174 1423
DFBAD1	7526	3032	3024
DFCDF	7462	2766	2760
DFGET	7542	3057	2745 2762 3001 3004 3062
DFLOOP	7440	2744	3044
DFL100	7566	3111	3021
DFL377	7565	3110	2776
DFL70	7560	3103	2750 2755
DFL740	7564	3107	3101
DFMAD	7546	3063	2767
FR	7541	3056	2742
FR	7567	3112	3026
DrUPR	7507	3013	2743

SYMBOL	VALUE	DEF	REFERENCES
DFSC	6622	2654	1445
DFSE	6621	2651	1443
DFWRIT	7432	2736	1447 3016
DFXCDF	7562	3105	2737 2772 3003 3012
DFXX	7561	3104	2761
DF32	6630	1343	2770 3011 3027 3106
DIALBL	0000	0054	0330
DIALCO	0010	0053	2115
DIALNU	0010	0055	2114
DIALUN	0100	0052	2116
DIHAN1	0365	0041	2113
DIHAN2	0366	0042	0123
DIMA	6616	2655	0125
DIML	6615	2653	0406 3020 3032
DIOC	0345	0040	2756
DISK	6623	2652	0121
DLCA	6755	1051	3014
OLC	6732	1033	1156
OLDR	6733	1034	0135 1123
DLW	6735	1035	1266
DLWC	6753	1050	1160
DMAC	6626	2656	
DMAR	6603	2643	1473 3056
DMAW	6605	2644	
DMT	6646	2657	
DMNT	6743	1043	
DPB00T	7174	2113	2076
DRCA	6757	1052	
ORCDF	6653	1411	1405
DRCLRF	6752	1530	1451 1461
DRDA	6734	1037	
DRDC	6736	1040	0137 2562
DRDS	6741	1041	1173
DRERR	6714	1454	1444 1450
DRLOOP	6640	1376	1466
DRL300	6731	1475	
DRL370	6746	1514	1421
DRL70	6730	1474	1402
DRL740	6743	1511	1506
DRMAD	6745	1513	1414
DRMAR	6727	1473	1374
DRMWC	6744	1512	1436
DRPR	6703	1441	1375
DRREAD	6630	1366	1372
DRRTR	6735	1502	1377 1400 1407 1416 1417 1425 1430 1431 1432 1464 1465 1532
DRWC	6752	1047	1410 1426 1433 1507
DRWRIT	6632	1370	1371 1403 1422 1427 1440
DRXCDF	6733	1477	1406
DRXX	6732	1476	1415 1437 1500
DR1	6636	1374	1367
DSAC	5612	2646	
DSAVI	4354	0626	0437 0462
DSKC	6745	1044	1165 1206 2072
DSKE	6747	1046	1170 1203
DSKT	6746	1045	
DSYS	7341	2417	0700
DTABLE	7247	2242	2530
DX	6001	0753	2316 2525 2563
DXAC	6645	2662	0150 2771
DXAL	6643	2661	0146 0405 2752 3000

SYMBOL	VALUE	DEF	REFERENCES
IB1	4346	0604	0444 0445
IB2	4352	0614	0491 0453
ICHECK	4214	0404	0322 0411
IDONE	4222	0414	0323 0372
IIDONE	4262	0460	
IIDONR	4265	0463	0434
ILINP	4344	0602	0442
ILOOP	4244	0440	0455
ILOUTP	4350	0612	
IMOVE	4113	0217	0166 0231 0246 0661
INITB	0311	0037	
IRF08	4126	0243	0153
IRK08	4200	0357	0321
ISETB	4311	0523	0326 0363 0525 0526 0533
ISETD	4267	0473	0324 0357 0475 0476 0503
ISETS	4300	0507	0325 0361 0511 0512 0517
ISETU	4340	0561	0327 0365 0366 0367 0564
ISETU2	4322	0540	0555 0562 0563
KCOMM	6436	1100	1072
KCOUNT	6550	1235	1142 1221
KDLDOR	6565	1266	1100
KEND	6545	1221	1167
KERR	6506	1152	1143 1213
KFGET	6553	1245	1112 1124 1133 1136 1250
KLOC	6552	1237	1111 1122 1132 1145 1147 1155
KLPH10	6452	1114	1105
KL30	6560	1260	1107
KL400	6562	1262	1146
KL7	6557	1257	
KL70	6561	1261	1103 1106 1121
KL7400	6563	1263	1115 1130 1157
KL7777	6564	1264	1131
KNEXT	6502	1145	1222
KOPR	6516	1163	1101
KREAD	6430	1071	1075 1102 1117 1140 1223 1246 1247
KTRACK	6551	1236	1116 1134 1135 1150 1161
KUGH	6524	1173	1205
KWRITE	6432	1073	1074 1176 1210
LBLOCK	7612	3231	3233
LCDF	7136	2026	1574 1577
LESS20	7041	1632	1623
LLOC	7611	3230	3234
LNUM	7613	3232	
LTEN	4347	0605	0443 0452
LUNIT	7610	3227	
L20	7147	2037	1632
L30	7150	2040	1631
L6	7145	2035	2066
L7	7146	2036	1627
L7770	7151	2041	1740 1744
L7771	7152	2042	1743 1756
MAIN	7000	1571	3134 3534
MCALL	7144	2034	1616 1620 1747 1751 1752 1754 1760 1762 1763
MLOOP	7044	1640	1652
MNL	7400	2525	1675 2541 2550
MNOT	7057	1670	1642
MOVE	7200	2157	0331 2214 2222 2223
MOVE	7137	2027	1603 1610 2075
MOVE	7143	2033	1607 1615 1617 1777
MSYS	7330	2406	0662 0663 0664 0665 0666 0667

SYMBOL VALUE DEF REFERENCES

MSYSF	4364	0664	0516						
MSYSI	4365	0665	0514						
MSYSP7	4366	0666	0532						
MSYS10	4367	0667	0530						
MTABLE	7300	2356	2031						
MTBUE	7064	1734	1646						
MWRITE	7140	2030	1606	1611	2074				
M20	7153	2043	1621						
M30	7154	2044	1624						
M40	7155	2045	1630						
PTABLE	7141	2031	1612						
RCDF	7134	2020	1575	1774					
READ	7774	3536	0112	0672					
RF08	7430	2712	0332						
RK08	6430	1007	0660						
RK080K	7165	2074	2070						
SETUP1	0322	0035	0421	2307					
SETUP2	0323	0036	0423	2311					
SLCDF	7246	2234	2172						
SRCDF	7133	2017	1614	1746	1757	2021			
STCDF	7130	2011	1602	1634	1755	1765	2013		
SWGET	7236	2221	2161	2163	2165	2174	2200	2211	2224
SWIN	7222	2201	2162	2212					
SWLIN	7244	2232	2164	2202	2207				
SWLOUT	7245	2233	2170	2204	2205				
SWOUT	7224	2203	2166						
SWRET	7234	2213	2173	2176					
SWTMP1	7242	2230							
SWTMP2	7243	2231							
SYSBOT	7777	3541	0675						
SYSGO	7617	3237	3221						
SYSLD2	7615	3234	3225						
SYSLD	7600	3217	3235						
TAXO	7727	3435	3371	3404	3406				
TBLOCK	7732	3441	3417	3453					
TCALLY	7750	3465	3360	3376	3377	3407	3415	3416	3420 3421
TCDF	7131	2012	1600						
TCOUNT	7751	3466	3425	3454					
TDO	7653	3356	3323	3343	3361	3362	3473	3474	3475
TEMP	7142	2032	1613	1640	1650	1651	1734	1737	1741 1742 1745 1766
TL00P	7723	3430	3455						
TL160	7760	3507	3400						
TL20	7757	3506	3405						
TL400	7637	3327	3450						
TL7	7756	3505	3367						
TL7400	7761	3510	3413						
TOPR	7731	3440	3363	3374	3375				
TOUT	7752	3472	3424	3456					
TPOINT	7627	3266	3217	3223					
TREAD	7630	3316	3226	3322	3337	3340	3341	3357	
TRET	7762	3511	3365	3446	3447	3472			
TTREAD	7647	3343	3317	3347					
TTRET	7643	3336	3350						
TWRITE	7632	3320	3321	3334	3414	3430	3451	3452	
WRITE	7775	3537	0113	0673					

